Predicting Gaps in Humanitarian Response Funding

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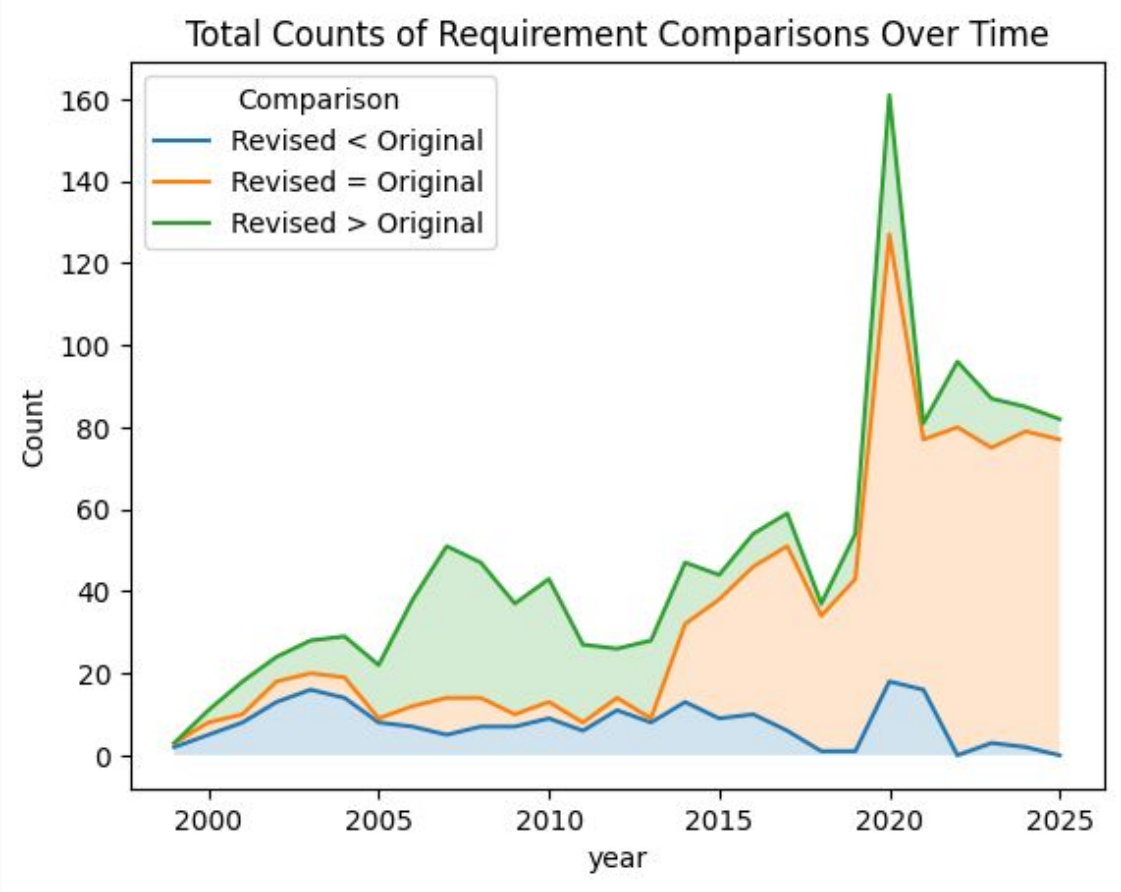
Problem

- Over the years, the UN has executed several humanitarian response plans to address issues around the world.
- We have a dataset which contains funding information for each of them.
- However, these plans do not match up exactly with actual humanitarian need.
- How can we identify these gaps between funding and needed and address them?

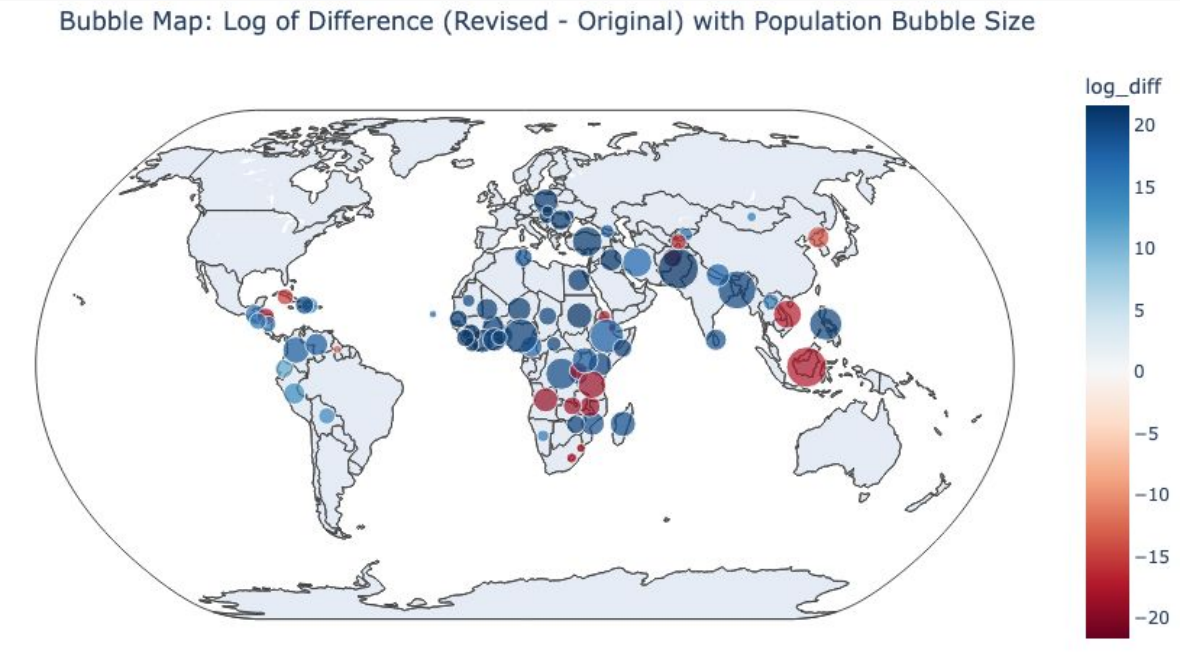
Our Goal

We want to predict whether a proposed amount of funding is accurate, too high, or too low for a humanitarian response plan. This would enable the UN to better predict how they should distribute funds more effectively.

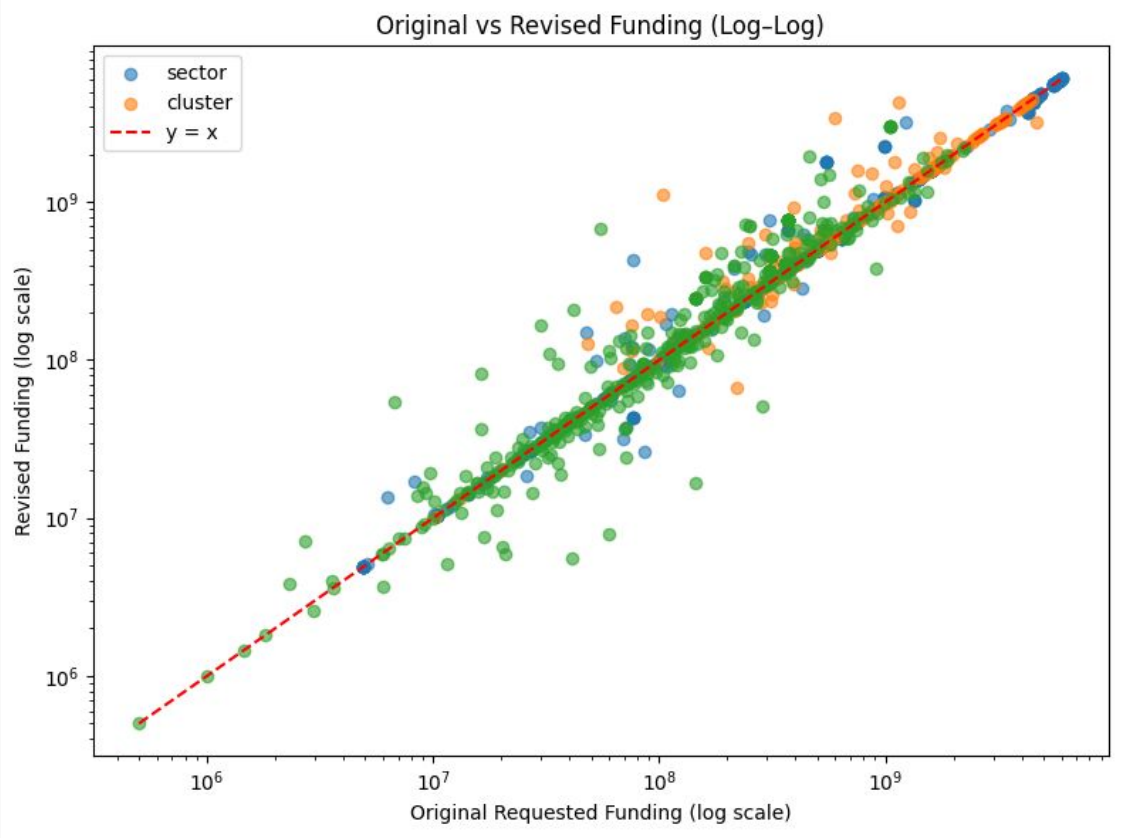
Gaps in Funding Over the Years



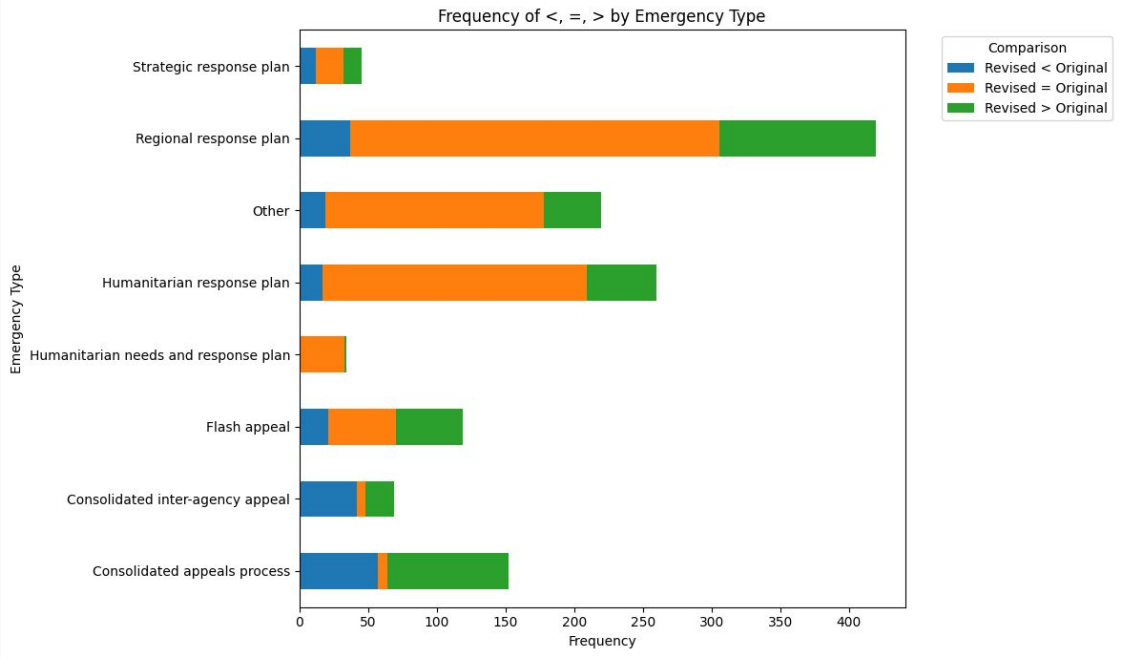
Gaps in Funding by Country



Original vs. Revised Funding By Response Type



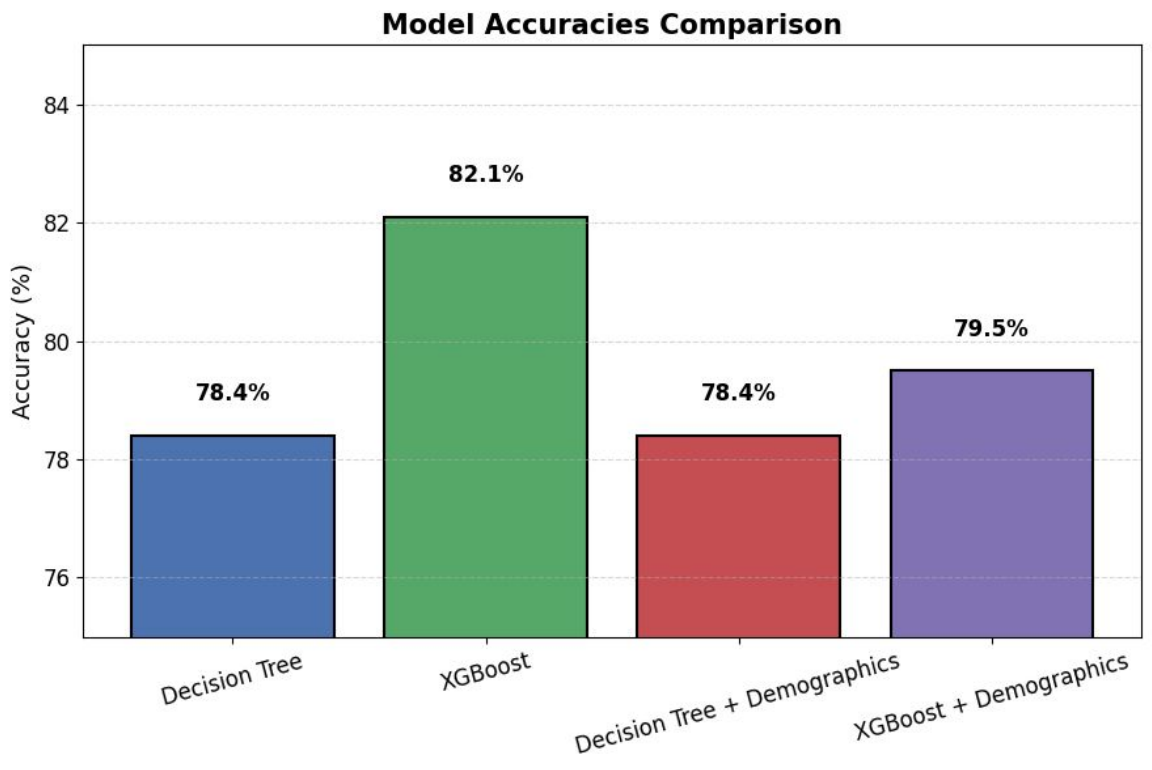
Gaps in Funding by Emergency type



Model

- Predict whether the revised funding requirement will be higher, lower, or equal to the original funding requirement.
- Model makes this prediction based on on original funding, type of emergency/response, country, and year.
- Later, we incorporated demographics like country total population and and total people in need.
- We tested a Decision Tree model for explainability and an XGBoost model for improved results.

Results


Model Accuracies Comparison
Accuracy (%)
84
82
80
78
76
82.1%
79.5%
78.4%
78.4%
Decision Tree
XGBoost
Decision Tree + Demographics
XGBoost + Demographics

Conclusions

- Budget revision behavior follows predictable trends

- Demographic data induced overfitting and may be redundant, since effect is captured within country

Proposed usages

- Budget revision is a time consuming process
- Our model can quickly determine if a budget revision is necessary
- Allows saved time, and more accurate revision judgment

Future improvements

- Source more accurate demographic data
- Source detailed humanitarian data to encode plan success
- More robust model tuning to avoid overfitting

Thanks!